

Background and Motivation

- Exponentially increasing satellite population in LEO
 - Starlink, Kuiper, etc.
- No current/past way to service satellites in orbit efficiently
 - NASA OSAM-1 cancelled
- Projected 50,000+ in LEO
- Inclination Target: [50, 71.6]°**
- Altitude Target: [500, 650] km**

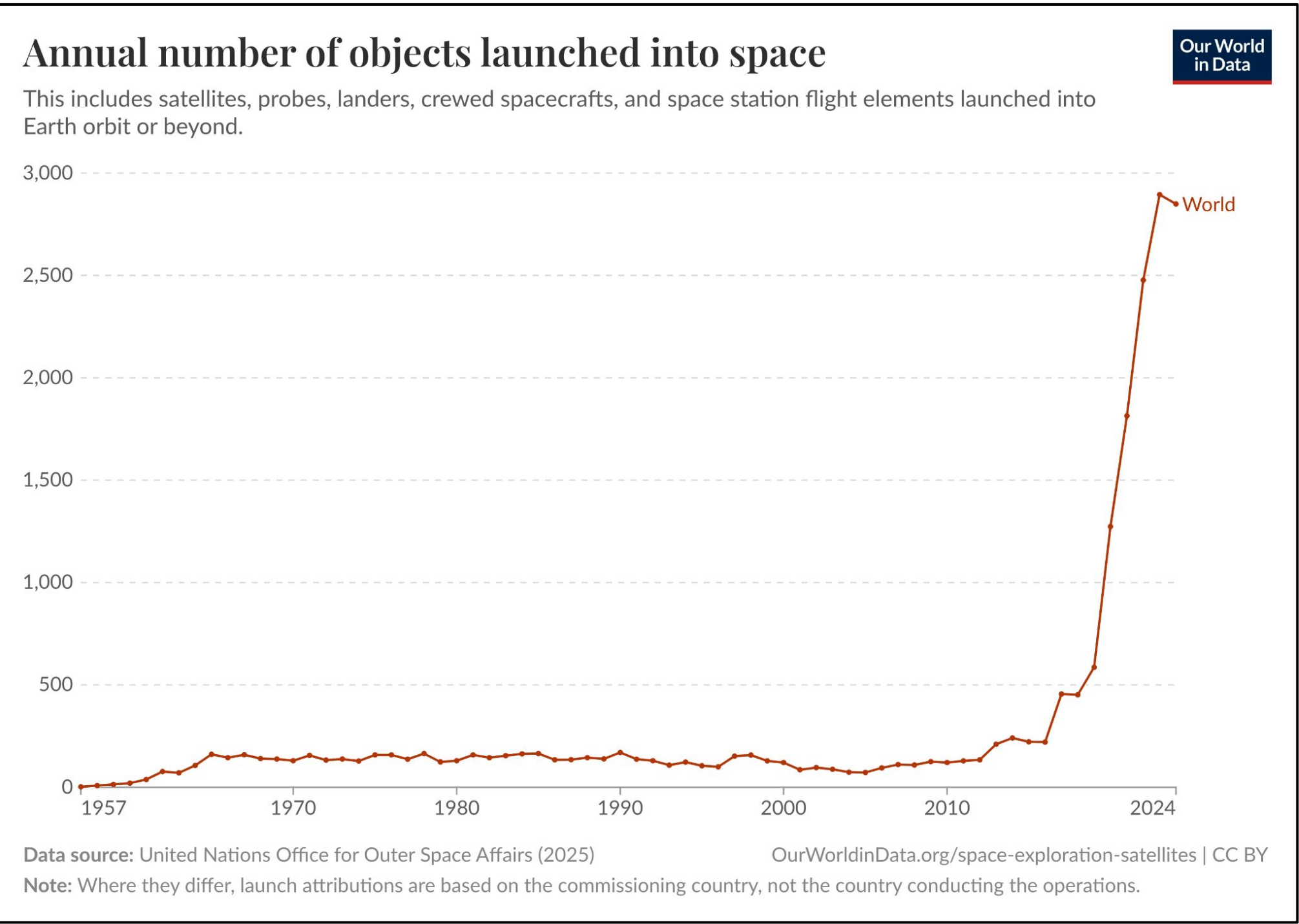


Figure 1. Annual Number of Object Launched into Space.

System Breakdown

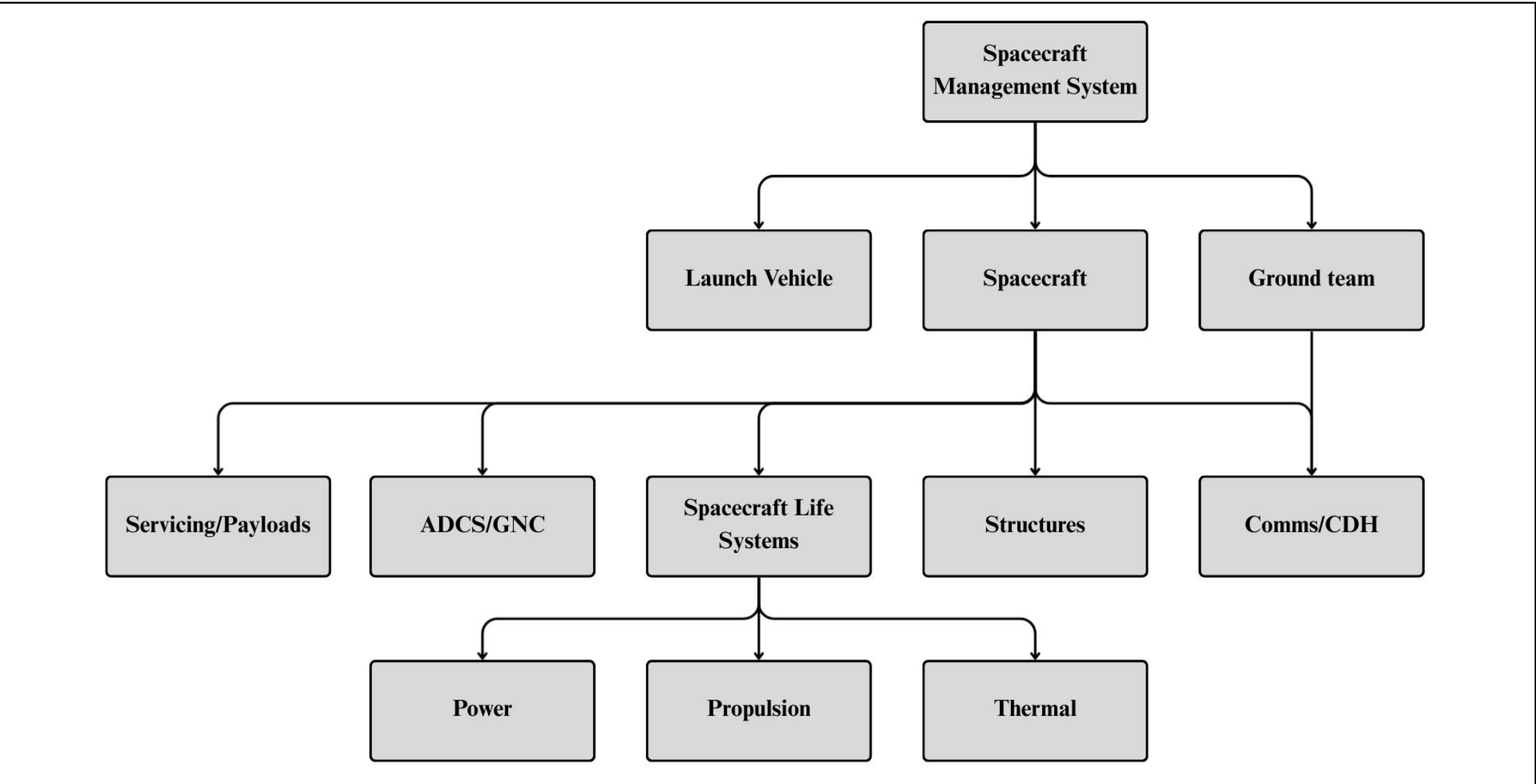


Figure 2. System Architecture.

What?

- SPARROW will perform *Critical Servicing Functions (CSF)* on client satellites
 - Extend lifetimes
 - Upgrade
 - Repair
 - Refuel
 - Data transfer
 - Orbital maneuvering

How?

- CSF will be performed by modular payloads
 - Hot-swappable
 - Customizable
 - Scalable
- Different clients require different payload
- Robotic arm(s)** = main payload

SPARROW Overview

- Fully autonomous and highly modular LEO servicing spacecraft for repair, refuel, and upgrades
- High power with 4000 m/s ΔV , capable of servicing multiple satellites in a single mission
- Reduces launches, costs, and orbital debris for sustainable space ops

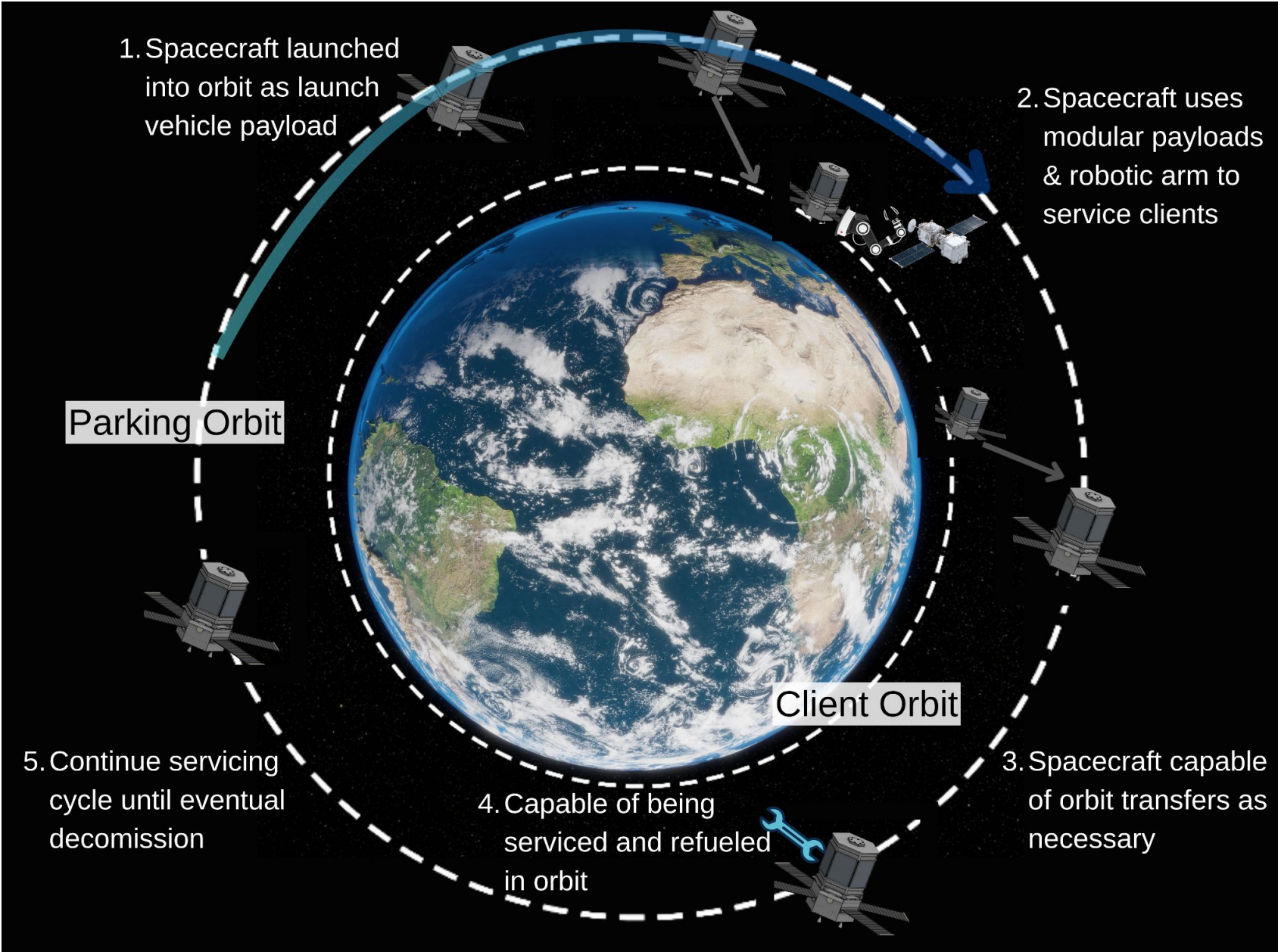


Figure 3. Concept of Operations.

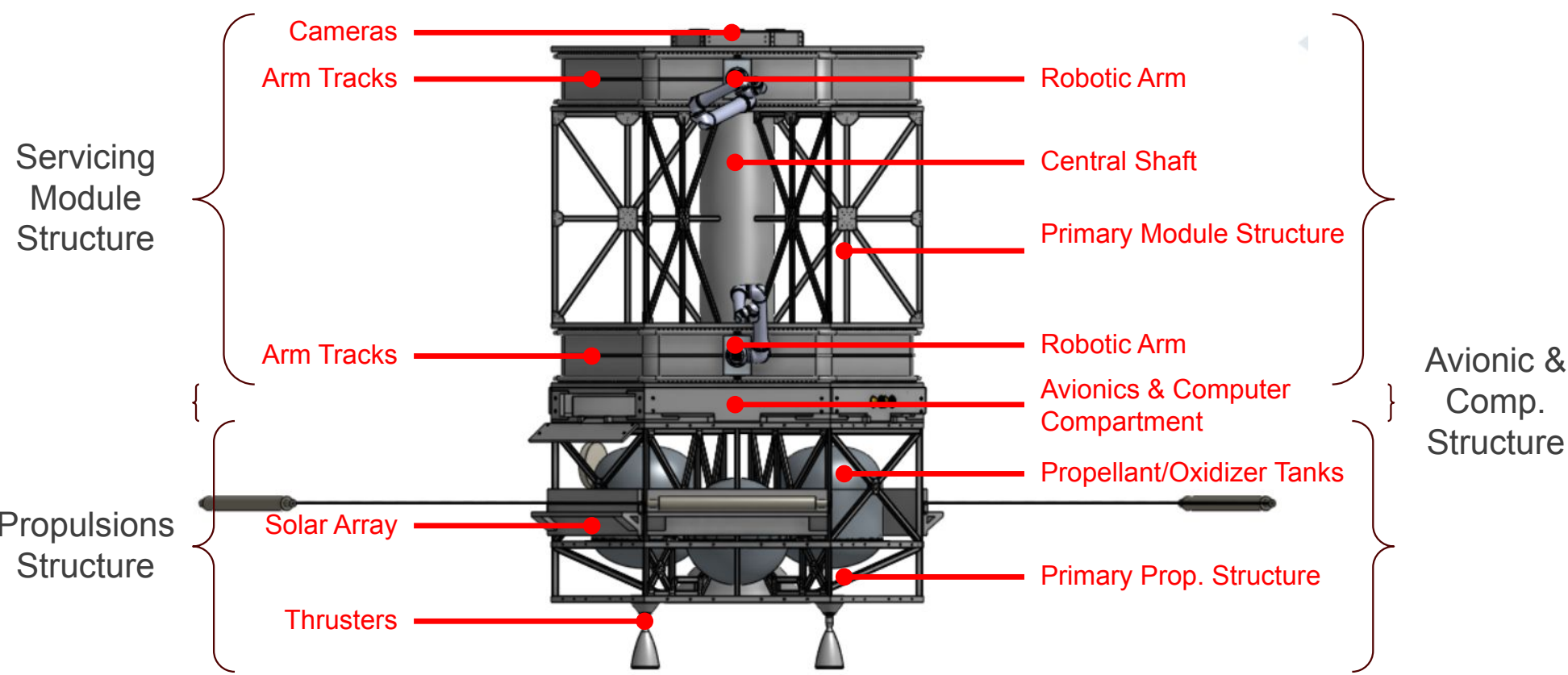


Figure 4. Main Spacecraft Bus.

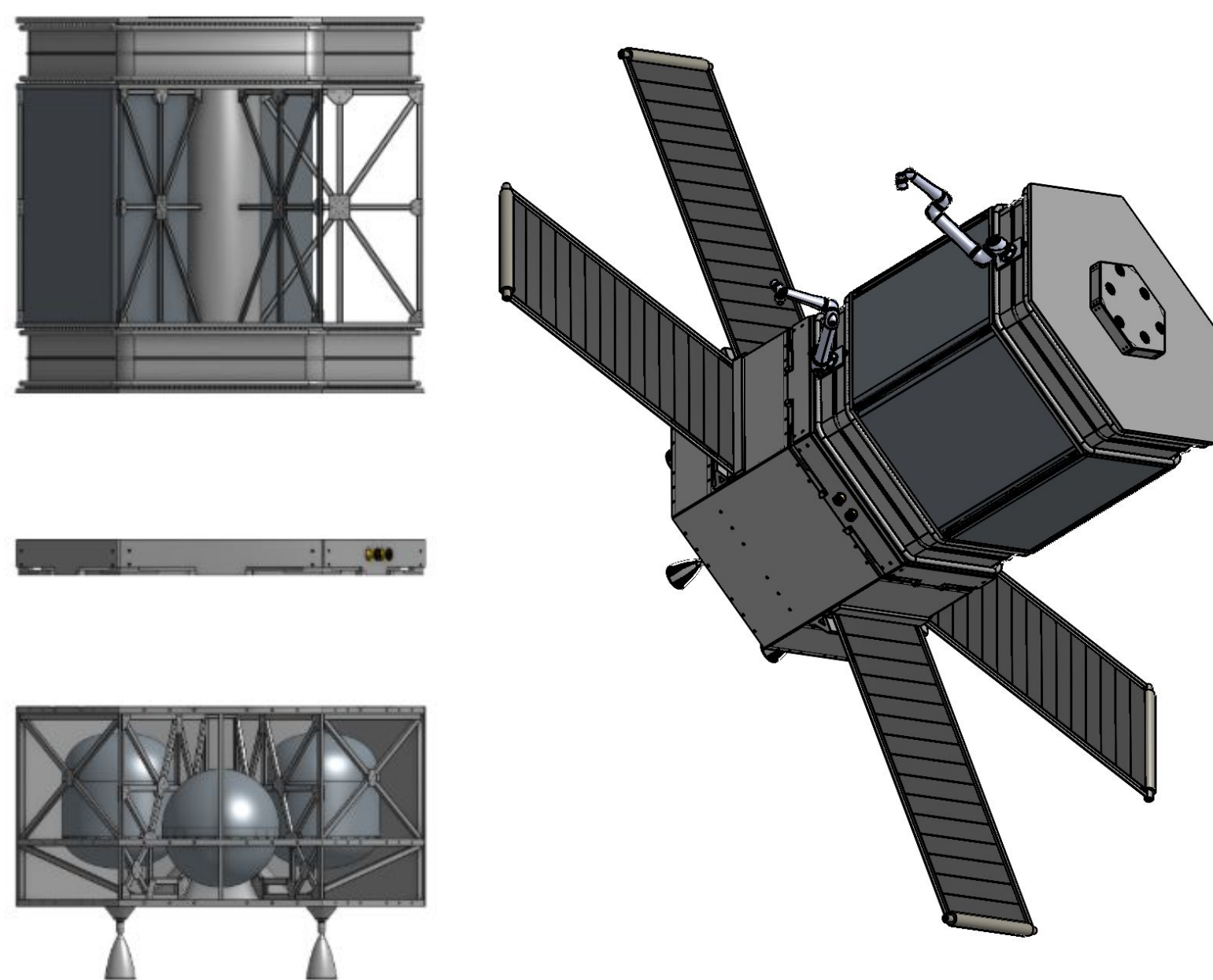


Figure 5. Sparrow Overview.

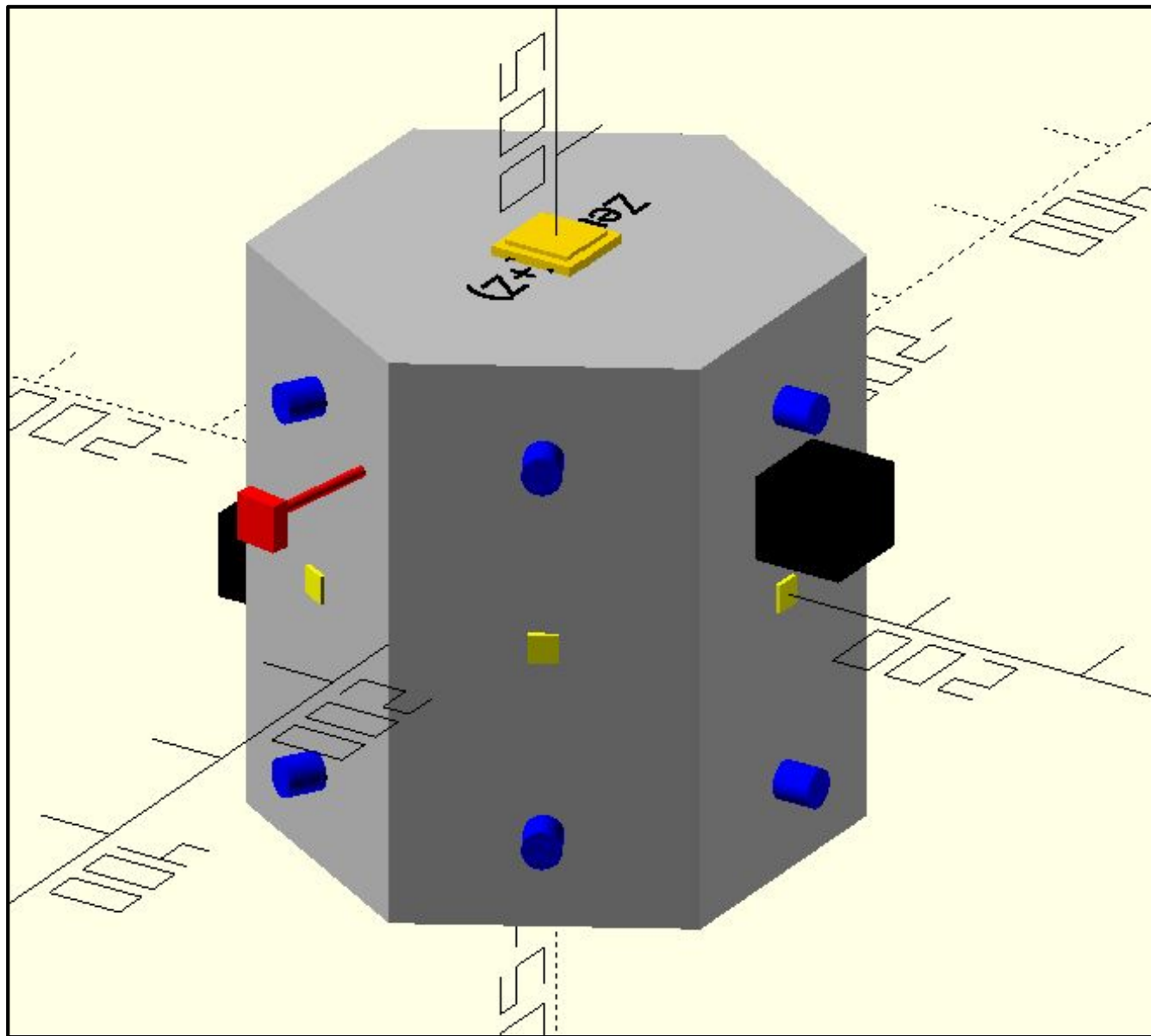


Figure 6. Sensor Placement.

Design Validation

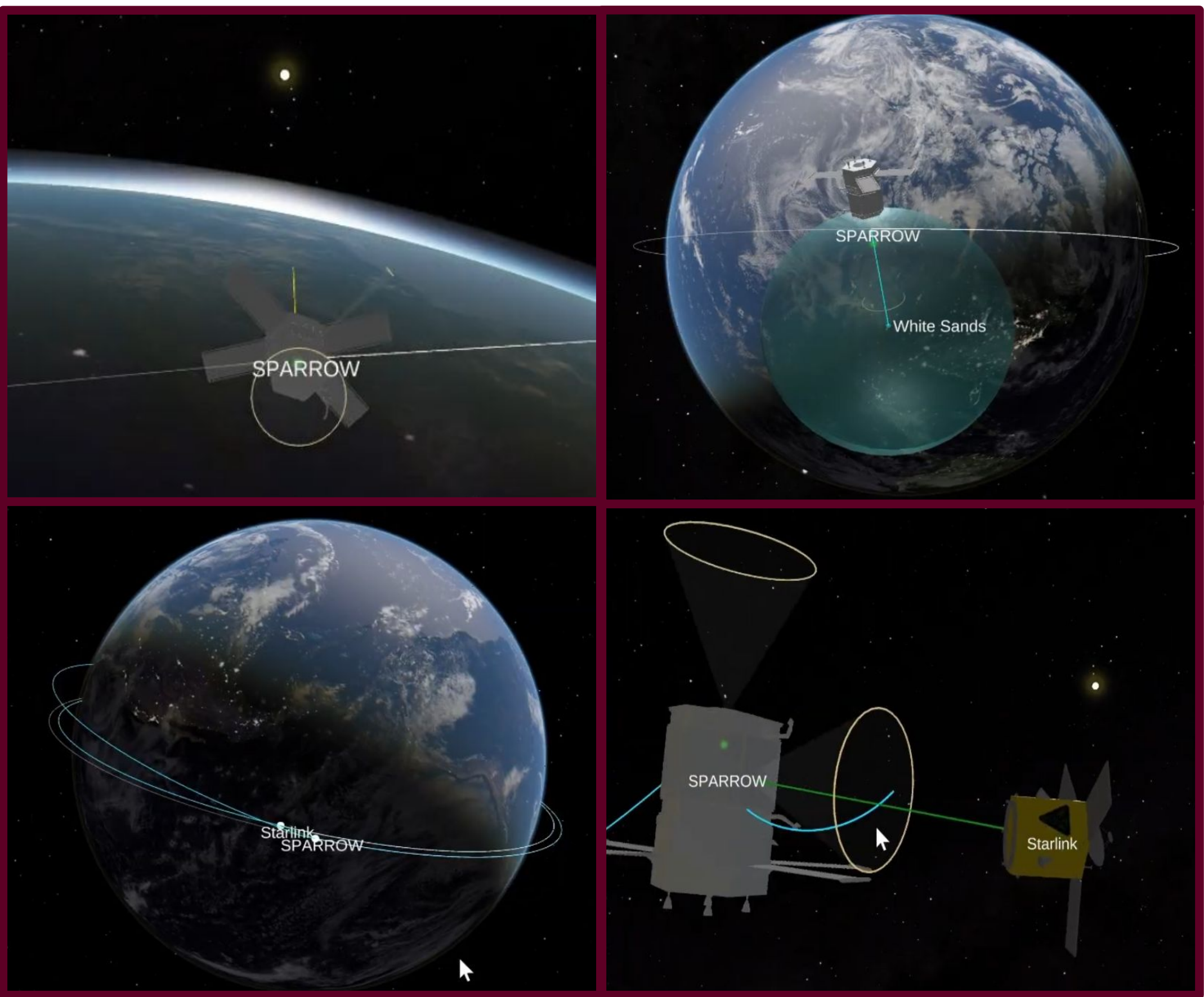


Figure 7. 4 Phase Validation Simulation.

We utilize astrodynamics simulation software **Basilisk** for SPARROW system validation.

- Truth Model:** 6-DOF Rigid Body Dynamics.
- Nav Model:** Corrupted by sensor noise specified in sensor datasheets.
- Power Model:** Solar panels, Battery configuration, component wise power usage, and eclipse detection.
- Mass Model:** Main thruster fuel consumption

Systems Validated:

- ADCS
- Power
- Propulsion
- Comms
- Thermal

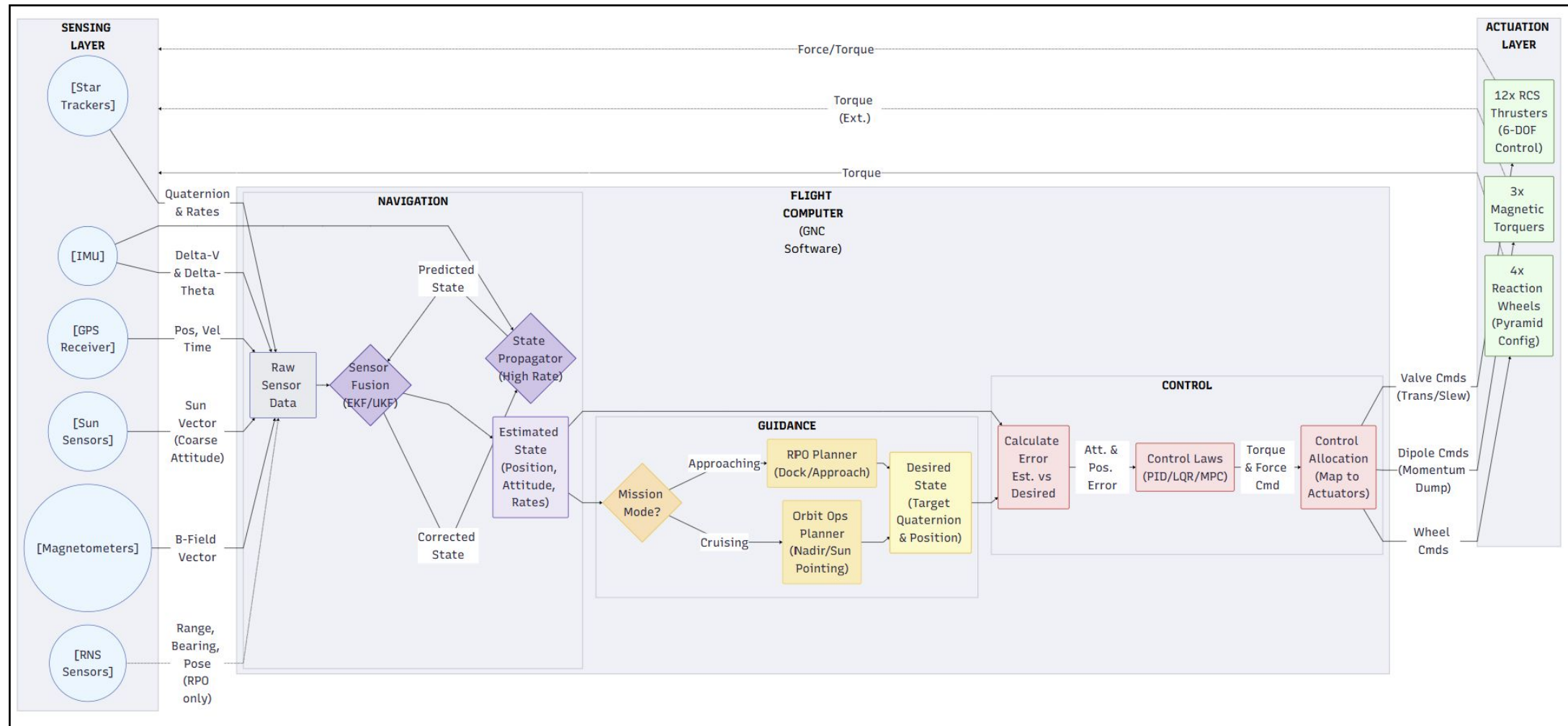


Figure 8. ADCS Control loop Architecture.

Innovative Concepts

- Robotic Arms on Circumferential Tracks (Pursued)
 - Allows for full, 360° authority around spacecraft
 - Able to access multiple servicing locations
 - Highly flexible for servicing
- Orbital Docking station (Not Pursued)
 - Gas station/garage in orbit with modular part swapping capability
 - Outside project scope, would increase complexity significantly
- Fully modular payload compartments (Pursued)
 - Clients can send modular compartments to SPARROW with needed tools/interfaces
 - Hot-swappable end effectors, tools, parts, etc.
 - Major driver of spacecraft architecture
 - Highly flexible and scalable
- Fully autonomous (Pursued)
 - Multiple on-board, specialized AI models
 - No human in the loop intervention
 - Allows scaling for satellite constellation

References

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Acknowledgements

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